



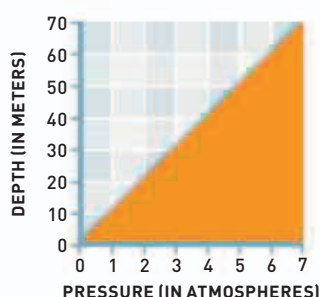
Simon Stevin wrote in Dutch, which he felt was better suited to technical subjects.

“ [THE DECIMAL FRACTION] TEACHES... THE EASY PERFORMANCE OF ALL RECKONINGS, COMPUTATIONS, AND ACCOUNTS, WITHOUT BROKEN NUMBERS. ”

Simon Stevin, Flemish mathematician and engineer, from *De Thiende* (The Tenth), 1585

#### FLEMISH MATHEMATICIAN AND ENGINEER SIMON STEVIN

(1548–1620) published the booklet *De Thiende* (The Tenth) in 1585. It promoted the use of decimal fractions and predicted the adoption of a decimal system of weights and measures. While



#### Water pressure

Directly proportional to depth, water pressure increases by 1 atmosphere (atm) for every 10m (33ft) in depth.

#### WATER PRESSURE

As an object descends into water, the weight of the water above it exerts pressure on it. As a result, water pressure increases with depth. At a depth of about 33 ft (10 m), water pressure is double the atmospheric pressure at the surface. Water pressure at ocean floors can be as much as 1,000 atmospheres; 1 atm equals 14 lb/in<sup>2</sup> (1 kg/cm<sup>2</sup>).

Islamic mathematicians had used decimal fractions centuries before, Stevin presented a comprehensive case for their use, citing ease of calculation. His notation was awkward and different from that used today.

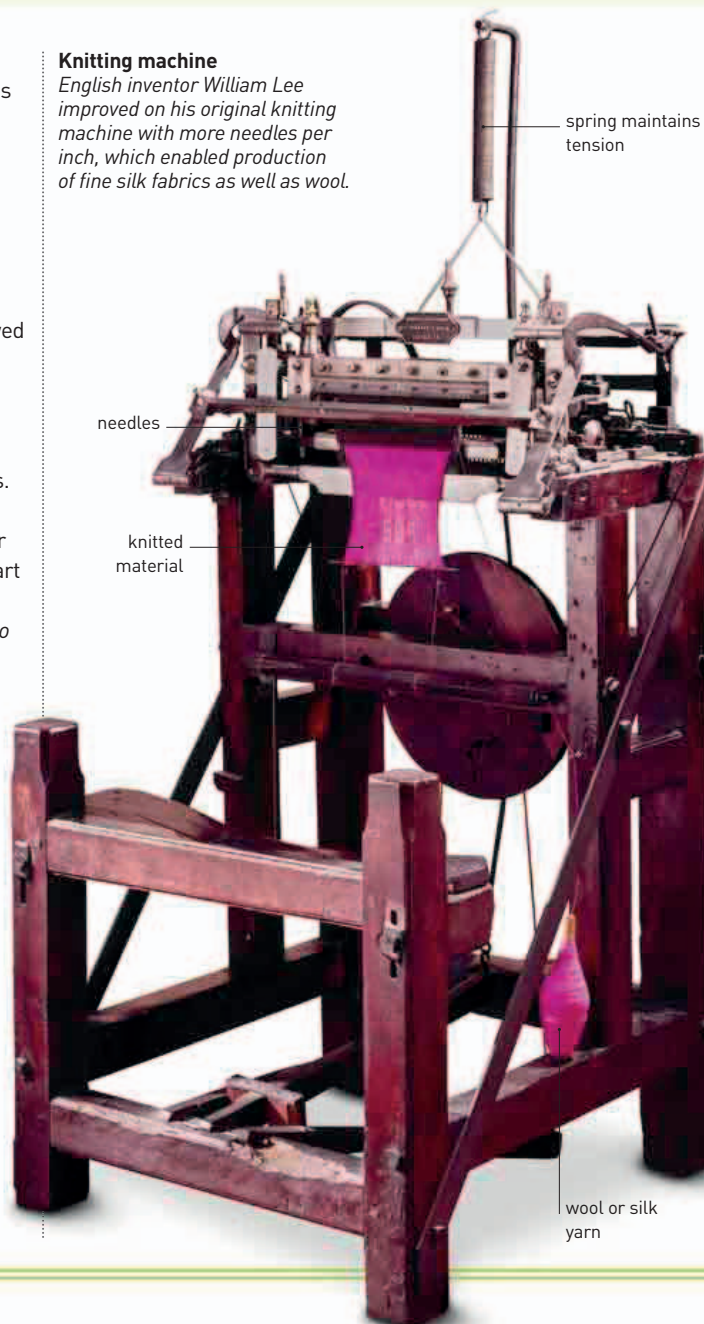
The following year, Stevin published two works on water and “statics,” in which he showed that because of its weight, **the pressure of water increases with depth**. His ideas became the foundation for a field of engineering called hydrostatics.

In 1588, Danish astronomer **Tycho Brahe** published further works, including the second part of his *Astronomiae Instauratae Progymnasmata* (Introduction to New Astronomy). He described the observation of comets and the instruments he used, and also included a catalog of stars, and described a **geo-heliocentric universe**, in which most of the planets orbited the Sun, and the Earth and Moon orbited the Earth.

In 1589, English inventor **William Lee** (1563–1614) designed the **stocking frame machine**, which mimicked the action of hand-knitters. Although it had the potential to revolutionize the textile industry, fear of upsetting the hand-knitters kept Lee from obtaining a patent in England, and he moved to France.

#### Knitting machine

English inventor William Lee improved on his original knitting machine with more needles per inch, which enabled production of fine silk fabrics as well as wool.



Galileo's thermoscope was an early device for measuring temperature.

#### DUTCH LENS-MAKER ZACHARIAS JANSEN

(1580–1638) is believed to have invented the microscope, initially using a single magnifying lens. Around 1590, he combined two lenses to form the **first compound optical microscope**, which was capable of magnifying images about nine times. Jansen is also associated with the development of the telescope, an invention credited to his rival Hans Lippershey in 1608.

With the publication of the ten-part *In Artem Analyticien Isagoge* (Introduction to the Art of Analysis) in 1591, French mathematician **François Viète**, also known as Vieta, laid the **foundations for modern algebra**. One of the key innovations of his system of analysis—known as “new algebra”—was the use of letters of the alphabet for parameters and unknowns in equations. Viète thereby created a symbolic algebra to replace the Classical and Islamic rhetorical algebra, which relied on explanation rather than signs and symbols.

In 1592, Italian mathematician **Galileo Galilei** invented the **thermoscope**, a tube in which liquid rises and falls with changes in temperature. This was the forerunner of the liquid thermometer, which was developed later by adding a scale to the tube.

1585 Simon Stevin publishes *De Thiende*, proposing a form of decimal fraction

1586 Stevin describes how water pressure increases with depth

1588 Tycho Brahe publishes two works on astronomy, including a catalog of stars

1589 William Lee builds a knitting machine

1590 Zacharias Jansen builds a compound microscope

1590 Galileo writes *De Motu*, an unfinished treatise on motion

1591 François Viète introduces “new algebra”

1592 Galileo invents the thermoscope